electricity). This fragmented approach creates challenges for unified analysis:

- *Swivel chair integration* - data is scattered across separate utility web portals, lacking a unified digital view.
- *Data format inconsistency* - historical data exists across varying formats (APIs, downloadable CSVs, and non-machine-readable PDF bills).
- *Infrequent updates* - quarterly billing prevents timely behavior modification.

To understand the 10-year consumption baseline (e.g., seasonal usage patterns) and feed historical data to machine learning models for prediction (e.g., testing the impact of weather and solar production), this locked PDF data must be digitized.

The solution is to use *optical character recognition (OCR)*, specifically an open-source library like Tesseract, to extract meter readings and dates from historical PDF bills. This converts the information into a historized, machine-readable format—although not perfect, OCR provides a valuable mechanism for digitizing paper records. The sample code in listing 2.1 uses Tesseract to extract the date and meter reading from the PDF bill shown in figure 2.1. Before running this code sample, ensure you have installed Tesseract (available from https://tesseract-ocr.github.io/tessdoc/Installation.html) on your system and that it is available in your path.

NOTE

All code samples are available in the book's GitHub repository here https://github.com/digital-twins-in-action, together with instructions on how to set up your Python environment to run them.